Production-Grade Kubernetes SRE Platform

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1 Production-Like Kubernetes SRE Lab Project

This lab is designed as a **portfolio-quality SRE / DevOps / Platform Engineering project** for:

- Senior SRE recruiters
- DevOps Engineer interviews
- Cloud Platform Engineer portfolios
- GitHub showcase projects

It simulates a realistic production Kubernetes environment locally using:

- k3d (recommended) or kind
- GitOps with Argo CD
- Monitoring with Prometheus + Grafana
- Logging with Loki + Promtail

- ingress-nginx
- HPA autoscaling
- Load testing
- Incident simulation
- Production-grade Kubernetes manifests

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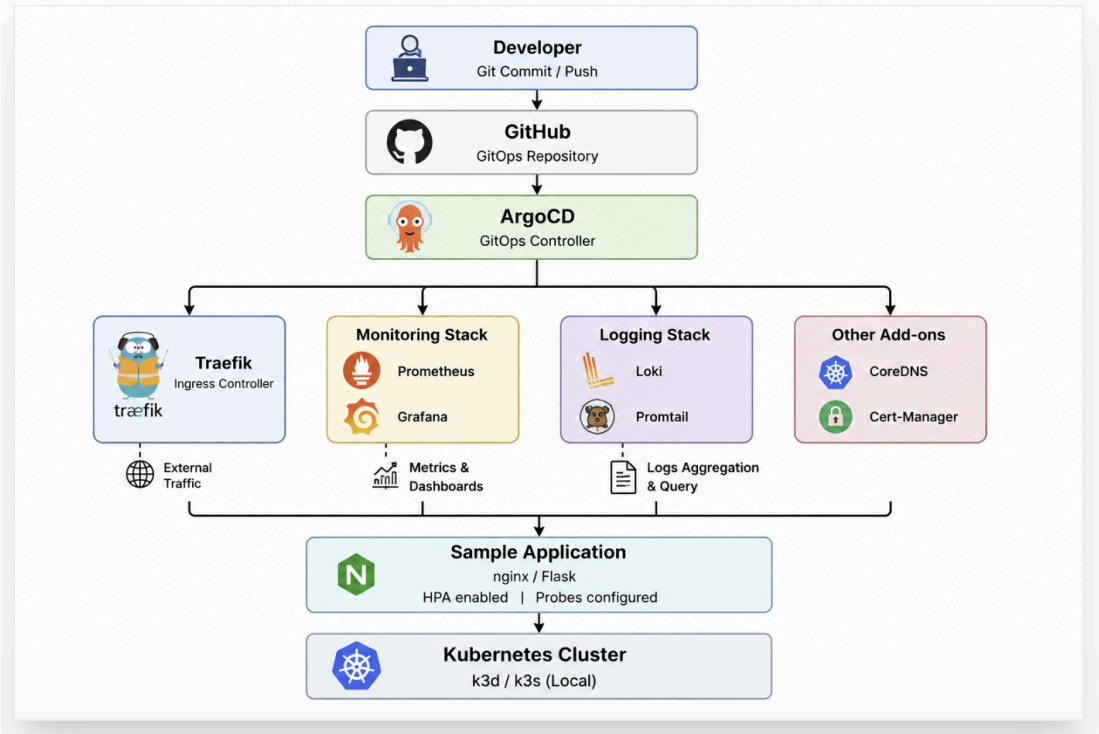
2. Production Architecture



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3. Recommended GitHub Repository Structure

3.1 Arborescence

```
henri@Henri:~/labs/sre-kubernetes-lab$ tree
.
├── README.md
├── app
│   ├── deployment.yaml
│   ├── hpa.yaml
│   ├── ingress.yaml
│   └── service.yaml
├── argo.test
├── argocd
│   └── application.yaml
├── cluster
│   └── k3d-config.yaml
├── docs
│   └── readme.md
├── incidents
├── ingress
├── load-testing
│   └── k6-loadtest.js
├── logging
│   ├── log-generator.yaml
│   ├── loki-stack.yaml
│   └── promtail-values.yaml
├── logs
│   ├── kukectk-logs-loki-0-logging-tail-100.log
│   └── log-loki-promtail-84cdl
├── monitoring
├── screenshots
│   └── readme.md
└── scripts

13 directories, 16 files
henri@Henri:~/labs/sre-kubernetes-lab$
```

3.3 Créer le dossier projet

mkdir sre-kubernetes-lab
cd sre-kubernetes-lab

3.4 creer la structure initiale :

mkdir -p \cluster \monitoring \logging \argocd \app \load-testing \incidents \scripts

3.5 initaliser git

Git init

3.6 créer readme minimal

touch README.md

3.7 premier commit

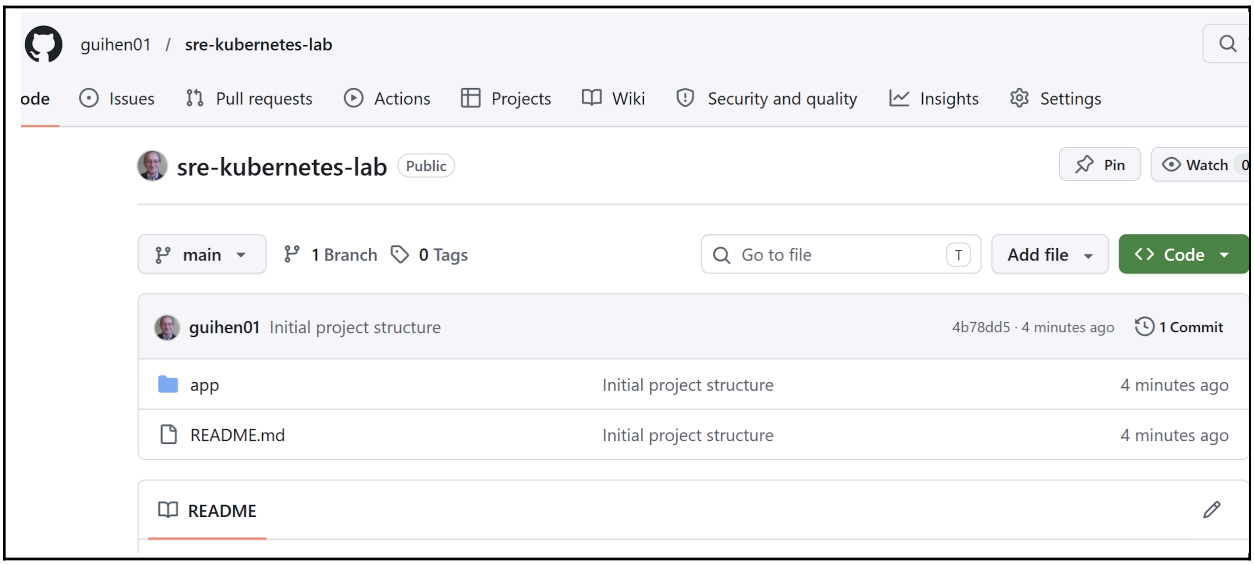
git add .

git commit -m "Initial project structure"

```
henri@Henri:~/labs/sre-kubernetes-lab$ pwd
/home/henri/labs/sre-kubernetes-lab
henri@Henri:~/labs/sre-kubernetes-lab$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/henri/labs/sre-kubernetes-lab/.git/
henri@Henri:~/labs/sre-kubernetes-lab$ touch README.md
henri@Henri:~/labs/sre-kubernetes-lab$ dir
README.md  app  argocd  cluster  incidents  ingress  load-testing  logging  monitoring  scripts
henri@Henri:~/labs/sre-kubernetes-lab$ git add .
henri@Henri:~/labs/sre-kubernetes-lab$ git commit -m "Initial project structure"
[master (root-commit) 4b78dd5] Initial project structure
 5 files changed, 92 insertions(+)
 create mode 100644 README.md
 create mode 100644 app/deployment.yaml
 create mode 100644 app/hpa.yaml
 create mode 100644 app/ingress.yaml
 create mode 100644 app/service.yaml
henri@Henri:~/labs/sre-kubernetes-lab$
```

3.8 Créer le repo github

Nom du repository : sre-kubernetes-lab
Se connecter sur votre compte github
Créer le repository :



3.9 Connecter le repo

git remote add origin https://github.com/<username>/sre-kubernetes-lab.git

3.10 premier push

git branch -M main
git push -u origin main

4. Incremental Implementation Phases

4.1 Phase 1 — Cluster + Ingress

4.1.1 Install k3d

curl -s https://raw.githubusercontent.com/k3d-io/k3d/main/install.sh | bash

4.1.2 Create Cluster

```
k3d cluster create sre-lab \
  --agents 2 \
  -p "8080:80@loadbalancer" \
  -p "8443:443@loadbalancer"
```

Autre option pour créer le cluster : k3d cluster create --config cluster/k3d-config.yaml

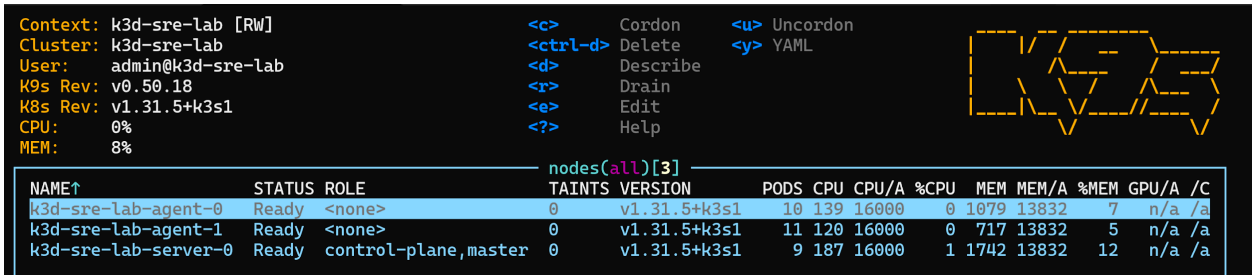
Cela créera 1 pod de type server et 2 pods de type workers

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get nodes
NAME                     STATUS   ROLES                  AGE   VERSION
k3d-sre-lab-agent-0      Ready    <none>                 80s   v1.31.5+k3s1
k3d-sre-lab-agent-1      Ready    <none>                 80s   v1.31.5+k3s1
k3d-sre-lab-server-0     Ready    control-plane,master   84s   v1.31.5+k3s1
```

En vérifiant les clusters, on vois que le nouveau cluster a 1 pod server (1/1) qui est présentement actif et 2 pods de type workers (2/2) qui sont actifs.

```
henri@Henri:~/labs/sre-kubernetes-lab$ k3d cluster list
NAME          SERVERS   AGENTS   LOADBALANCER
gitops-lab    0/1       0/0      true
henri         0/1       0/0      true
ingress-lab   0/1       0/0      true
platform-lab  0/1       0/0      true
sre-lab       1/1       2/2      true
henri@Henri:~/labs/sre-kubernetes-lab$
```

Avec k9s :



4.1.3 Traefik or gninx-ingress

If traefik no installed then Install ingress-nginx (seulement si trafik n'est pas installé)

kubectl apply -f https://raw.githubusercontent.com/kubernetes/ingress-nginx/main/deploy/static/provider/cloud/deploy.yaml

Verify: kubectl get pods -n ingress-nginx

Else

kubectl get pods -n kube-system

on vérifie que traefik est installé. Ce qui est le cas dans notre lab.

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n kube-system
NAME                                      READY   STATUS    RESTARTS      AGE
coredns-ccb96694c-94jd6                   1/1     Running   1 (15m ago)   14h
local-path-provisioner-5cf85fd84d-2dxrt   1/1     Running   1 (15m ago)   14h
metrics-server-5985cbc9d7-fmh7c           1/1     Running   1 (15m ago)   14h
svclb-traefik-e0711a08-6sbg2              2/2     Running   8 (15m ago)   3d
traefik-5d45fc8cc9-bl7jg                  1/1     Running   1 (15m ago)   14h
henri@Henri:~/labs/sre-kubernetes-lab$
```

Vérifier les IngressClass :

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get ingressclass
NAME      CONTROLLER                      PARAMETERS   AGE
traefik   traefik.io/ingress-controller   <none>       3d1h
henri@Henri:~/labs/sre-kubernetes-lab$
```

endif

4.1.4 Créer le namespace :

kubectl create namespace demo-app

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get ns
NAME              STATUS   AGE
default           Active   33m
demo-app          Active   8s
kube-node-lease   Active   33m
kube-public       Active   33m
kube-system       Active   33m
henri@Henri:~/labs/sre-kubernetes-lab$
```

4.2 Phase 2 — Monitoring Stack

Add Helm Repositories

helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm repo update

Install kube-prometheus-stack

helm install monitoring prometheus-community/kube-prometheus-stack \
  --namespace monitoring –create-namespace

Verify:

kubectl get pods -n monitoring

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n monitoring
NAME                                                     READY   STATUS    RESTARTS   AGE
alertmanager-monitoring-kube-prometheus-alertmanager-0   2/2     Running   0          24s
monitoring-grafana-74598f477b-c8sfj                      2/3     Running   0          28s
monitoring-kube-prometheus-operator-6c8c5bb9b-vmz7d      1/1     Running   0          28s
monitoring-kube-state-metrics-64485fd97-8bmjv            1/1     Running   0          28s
monitoring-prometheus-node-exporter-hvdc8                1/1     Running   0          28s
monitoring-prometheus-node-exporter-mxplk                1/1     Running   0          28s
monitoring-prometheus-node-exporter-qkvkk                1/1     Running   0          28s
prometheus-monitoring-kube-prometheus-prometheus-0       2/2     Running   0          24s
henri@Henri:~/labs/sre-kubernetes-lab$
```

4.3 Phase 3 — Logging Stack

Install Loki Stack

helm repo add grafana https://grafana.github.io/helm-charts
helm repo update

helm install loki grafana/loki-stack \
  --namespace logging \
  --create-namespace

Verify:

kubectl get pods -n logging

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get ns
NAME              STATUS   AGE
default           Active   44m
demo-app          Active   11m
kube-node-lease   Active   44m
kube-public       Active   44m
kube-system       Active   44m
logging           Active   16s
monitoring        Active   3m15s
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n logging
NAME                  READY   STATUS    RESTARTS   AGE
loki-0                0/1     Running   0          30s
loki-promtail-84cdl   1/1     Running   0          30s
loki-promtail-sq7r2   1/1     Running   0          30s
loki-promtail-tvt9v   1/1     Running   0          30s
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$
```

kubectl get svc -n monitoring

```
HXSaAKWE0TDjsT0yhenri@Henri:~/labs/sre-kubernetes-lab$ kubectl get svc -n monitoring
NAME                                     TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)                      AGE
alertmanager-operated                    ClusterIP   None            <none>        9093/TCP,9094/TCP,9094/UDP   21m
monitoring-grafana                       ClusterIP   10.43.232.73    <none>        80/TCP                       21m
monitoring-kube-prometheus-alertmanager  ClusterIP   10.43.4.217     <none>        9093/TCP,8080/TCP            21m
monitoring-kube-prometheus-operator      ClusterIP   10.43.109.93    <none>        443/TCP                      21m
monitoring-kube-prometheus-prometheus    ClusterIP   10.43.5.194     <none>        9090/TCP,8080/TCP            21m
monitoring-kube-state-metrics            ClusterIP   10.43.122.255   <none>        8080/TCP                     21m
monitoring-prometheus-node-exporter      ClusterIP   10.43.73.242    <none>        9100/TCP                     21m
prometheus-operated                      ClusterIP   None            <none>        9090/TCP                     21m
henri@Henri:~/labs/sre-kubernetes-lab$
```

4.4 Phase 4 — Tester Grafana

kubectl port-forward svc/monitoring-grafana \
-n monitoring 3000:80

Pour récupérer le mot de passe admin de Grafana installé avec le chart kube-prometheus-stack :

```
kubectl get secret -n monitoring monitoring-grafana \
  -o jsonpath="{.data.admin-password}" | base64 --decode ; echo
```

http://localhost:3000

4.5 Phase 5— Tester Prometheus

kubectl port-forward svc/monitoring-kube-prometheus-prometheus \
-n monitoring 9090

4.6 Phase 6 — GitOps with ArgoCD

Install ArgoCD :

kubectl create namespace argocd
kubectl apply -n argocd \
-f https://raw.githubusercontent.com/argoproj/argo-cd/stable/manifests/install.yaml

Expose UI:

kubectl port-forward svc/argocd-server -n argocd 8081:443

Get password:

kubectl -n argocd \
get secret argocd-initial-admin-secret \
-o jsonpath="{.data.password}" | base64 -d

Vérifier :

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n argocd
NAME                                                READY   STATUS    RESTARTS        AGE
argocd-application-controller-0                     1/1     Running   0               4h12m
argocd-applicationset-controller-866cbd67bd-72n4g   1/1     Running   19 (2m2s ago)   4h12m
argocd-dex-server-cdd7bd88f-p8wdr                   1/1     Running   0               4h12m
argocd-notifications-controller-5b59cc856f-xnhqm    1/1     Running   0               4h12m
argocd-redis-548bf6d4d7-snjkh                       1/1     Running   0               4h12m
argocd-repo-server-869988b885-f96np                 1/1     Running   0               4h12m
argocd-server-56b6d45cd6-fg2n9                      1/1     Running   0               4h12m
henri@Henri:~/labs/sre-kubernetes-lab$
```

---

4. Phase 7 — Autoscaling + Load Testing

Install Metrics Server

kubectl apply -f \
https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml

Vérifier le serveur de métriques :

kubectl get deployment metrics-server -n kube-system

```
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get deployment metrics-server -n kube-system
NAME             READY   UP-TO-DATE   AVAILABLE   AGE
metrics-server   1/1     1            1           7h41m
henri@Henri:~/labs/sre-kubernetes-lab$
```

On dois voir quelque chose comme : metrics-server-xxxxx Running

kubectl get apiservices | grep metrics

```
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get apiservices | grep metrics
v1beta1.metrics.k8s.io                 kube-system/metrics-server   True        7h42m
henri@Henri:~/labs/sre-kubernetes-lab$
```

on dois voir : metrics.k8s.io et True
exemple : v1beta1.metrics.k8s.io kube-system/metrics-server True

vérifier les métriques nodes :

kubectl top nodes

On dois obtenir :
NAME CPU(cores) CPU% MEMORY(bytes)
k3d-sre-lab-agent 120m 6% 800Mi

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl top nodes
NAME                     CPU(cores)   CPU(%)   MEMORY(bytes)   MEMORY(%)
k3d-sre-lab-agent-0      134m         0%       1094Mi          7%
k3d-sre-lab-agent-1      112m         0%       678Mi           4%
k3d-sre-lab-server-0     175m         1%       1746Mi          12%
henri@Henri:~/labs/sre-kubernetes-lab$
```

vérifier le métriques pods :

kubectl top pods -A

on dois voir CPU/MEM des pods.

---

5 Deployment

5.1 App/deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: demo-app
  namespace: demo-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: demo-app
  template:
    metadata:
      labels:
        app: demo-app
    spec:
      containers:
      - name: nginx
        image: nginx:stable
        ports:
        - containerPort: 80
```

```
    resources:
      requests:
        cpu: "100m"
        memory: "128Mi"
      limits:
        cpu: "500m"
        memory: "512Mi"

    livenessProbe:
      httpGet:
        path: /
        port: 80
      initialDelaySeconds: 10
      periodSeconds: 5

    readinessProbe:
      httpGet:
        path: /
        port: 80
      initialDelaySeconds: 5
      periodSeconds: 3
```

5.1.1 Deployer application :

kubectl apply -f app/deployment.yaml

5.1.2 verifier pods :

kubectl get pods -n demo-app

```
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl apply -f app/deployment.yaml
deployment.apps/demo-app created
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n demo-app
NAME                        READY   STATUS    RESTARTS   AGE
demo-app-58787c9f9d-5xjz6   1/1     Running   0          20s
demo-app-58787c9f9d-vtqgd   1/1     Running   0          20s
henri@Henri:~/labs/sre-kubernetes-lab$
```

5.2 App/service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: demo-app
  namespace: demo-app
```

```
spec:
  selector:
    app: demo-app
  ports:
  - port: 80
    targetPort: 80
```

5.2.1 Déployer service

kubectl apply -f app/service.yaml

5.2.2 verifier service :

kubectl get svc -n demo-app

```
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl apply -f app/service.yaml
service/demo-app created
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get svc -n demo-app
NAME       TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
demo-app   ClusterIP   10.43.79.172   <none>        80/TCP    9s
henri@Henri:~/labs/sre-kubernetes-lab$
```

5.3 App/ingress.yaml

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: demo-app
  namespace: demo-app
spec:
  ingressClassName: traefik
  rules:
  - host: demo.local
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: demo-app
            port:
              number: 80
```

5.3.1 Déployer ingress

kubectl apply -f app/ingress.yaml

5.3.2 verifier ingress

kubectl get ingress -n demo-app

```
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl apply -f app/ingress.yaml
ingress.networking.k8s.io/demo-app created
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get ingress -n demo-app
NAME       CLASS     HOSTS        ADDRESS                              PORTS   AGE
demo-app   traefik   demo.local   172.21.0.3,172.21.0.4,172.21.0.5   80      22s
henri@Henri:~/labs/sre-kubernetes-lab$
```

5.4 App/hpa.yaml

5.4.1 Créer le fichier app/hpa.yaml

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: demo-app-hpa
  namespace: demo-app
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: demo-app

  minReplicas: 2
  maxReplicas: 10

    behavior:
    scaleDown:
      stabilizationWindowSeconds: 30

      policies:
      - type: Percent
        value: 50
        periodSeconds: 15

  metrics:
    - type: Resource
    resource:
      name: cpu
      target:
```

```
    type: Utilization

    averageUtilization: 5
```

5.4.2 Configuration du scale down

Analyse de la partie de code, liée au scale down , dans le fichier précédent : hpa.yaml :

```
behavior:
scaleDown:
stabilizationWindowSeconds: 30
policies:
- type: Percent
value: 50
periodSeconds: 15
```

Ici :
Kubernetes attend 30 s
puis peut supprimer maximum 50% des pods toutes les 15 s
Exemple :
10 pods
après 30 s ⟶ passe à 5
15 s plus tard ⟶ 3
puis 2

Cette partie de code ets importante, car au début elle, n'était pas présente et le cluster ne faisait pas vraiment de scale down visible, le nombre de pods restaient au maximum de 10

5.4.3 Déploiement du .yaml :

kubectl apply -f app/hpa.yaml

5.4.4 Vérifier HPA :

kubectl get hpa -n demo-app

```
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl apply -f app/hpa.yaml
horizontalpodautoscaler.autoscaling/demo-app-hpa created
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get hpa -n demo-app
NAME           REFERENCE              TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
demo-app-hpa   Deployment/demo-app    cpu: 1%/50%   2         10        2          14s
henri@Henri:~/labs/sre-kubernetes-lab$
```

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get hpa -n demo-app
NAME           REFERENCE              TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
demo-app-hpa   Deployment/demo-app    cpu: 1%/50%   2         10        2          21h
henri@Henri:~/labs/sre-kubernetes-lab$
```

5.4.5 tester application :

curl http://localhost:8080
OU BIEN
curl -H "Host: demo.local" http://localhost:8080

```
henri@Henri:/mnt/c/Users/henri$ curl -H "Host: demo.local" http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
```

5.4.6 Vérifier probes :

kubectl describe pod -n demo-app <pod-name>

chercher :
Liveness probe succeeded
Readiness probe succeeded

```
henri@Henri:/mnt/c/Users/henri$ kubectl get pods -n demo-app
NAME                        READY   STATUS    RESTARTS   AGE
demo-app-58787c9f9d-5xjz6   1/1     Running   0          10m
demo-app-58787c9f9d-vtqgd   1/1     Running   0          10m
henri@Henri:/mnt/c/Users/henri$
henri@Henri:/mnt/c/Users/henri$ kubectl describe pod -n demo-app demo-app-58787c9f9d-5xjz6
Name:             demo-app-58787c9f9d-5xjz6
Namespace:        demo-app
Priority:         0
Service Account:  default
Node:             k3d-sre-lab-agent-0/172.21.0.4
Start Time:       Fri, 08 May 2026 19:35:28 -0400
Labels:           app=demo-app
                  pod-template-hash=58787c9f9d
Annotations:      <none>
Status:           Running
```

5.5 commit git

faire un commit git afin que les fichiers locaux .yaml soient poussés vers le reposiroty github

git add .
git commit -m "Deploy stack"
git push

6. ArgoCD Application YAML

Argocd/application.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
```

```
  name: demo-app
  namespace: argocd

spec:
  project: default

  source:
    repoURL: https://github.com/yourname/sre-kubernetes-lab.git
    targetRevision: HEAD
    path: app

  destination:
    server: https://kubernetes.default.svc
    namespace: demo-app

  syncPolicy:
    automated:
      prune: true
      selfHeal: true
```

6.1 Vérification de la syntaxe

On vérifie la syntaxe du .yaml avant de le déployer. On utilise la commande pour faire un dry-run, une simulation de déploiement

kubectl apply --dry-run=client -f app/hpa.yaml

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl apply --dry-run=client -f app/hpa.yaml
horizontalpodautoscaler.autoscaling/demo-app-hpa configured (dry run)
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$
```

On dois voir :
horizontalpodautoscaler.autoscaling/demo-app-hpa created (dry run)

6.2 Deployer :

kubectl apply -f argocd/application.yaml

6.3 Vérification 1 :

kubectl get applications -n argocd

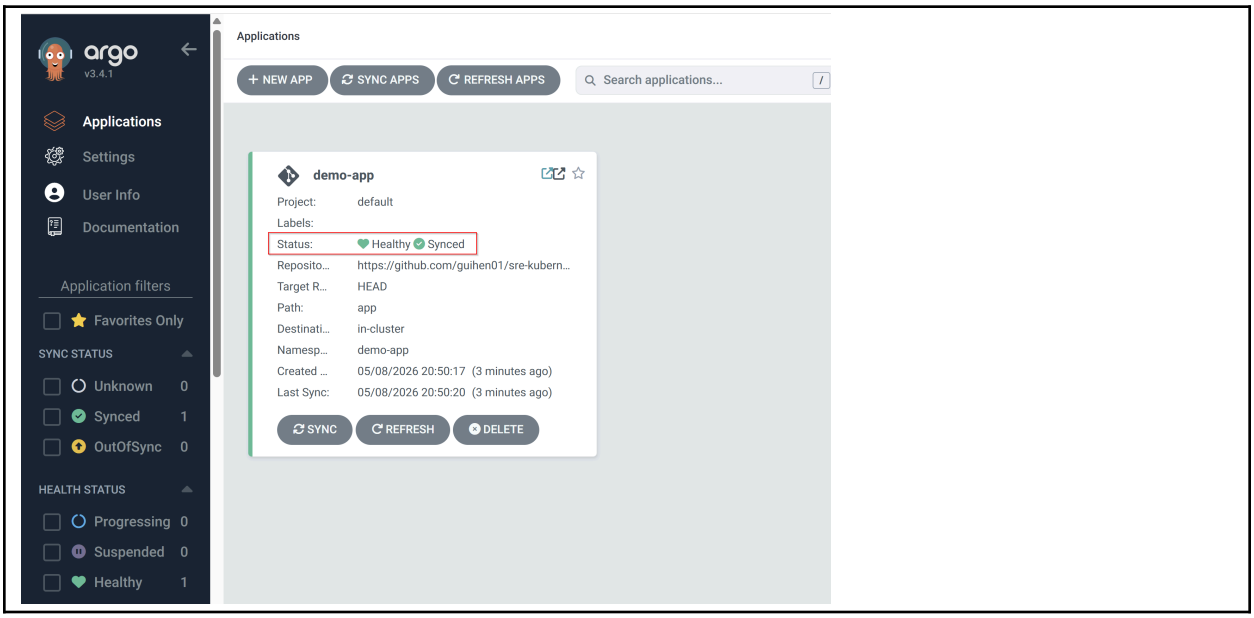
```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl apply -f argocd/application.yaml
application.argoproj.io/demo-app created
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get applications -n argocd
NAME       SYNC STATUS   HEALTH STATUS
demo-app   Synced        Healthy
henri@Henri:~/labs/sre-kubernetes-lab$
```

On dois obligatoirement voir s'afficher : Synced et Healthy

Si ce n'Est pas le cas, souvent la cause ets le .yaml mal configuré .
Réviser le .yaml (voir la commande au paragraphe eprécédent : 6.1)
Corriger la syntaxe .
Puis pousser vers le repo git, à nouveau.
git add .
git commit -m "Fix HPA YAML indentation"
git push
revérifier ensuite qu'on est à nouveau dans l'état synced

6.4 vérification 2 :

Puis : aller dans Argo UO : https://localhost:8081
Vérifier que : Healthy et Synced



6.5 Vérification en cours de lab de l'état de Argo

Toujours vérifier sir argo est dans l'état synced
Kubectl get applications -n argocd

On dois voir s'Afficher : synced healthy

Si ce n'est pas le cas , investiguer avec la commande :
kubectl describe application demo-app -n argocd

Vérifier dans la liste qui s'affiche les erreurs éventuelles
Corriger
Redeployer avec git

Si nécessaire , Forcer le refresh de argocd : 9efface la mémoire cache , force la réécriture

**kubectl annotate application demo-app **
**-n argocd **
argocd.argoproj.io/refresh=hard --overwrite

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl annotate application demo-app \
-n argocd \
argocd.argoproj.io/refresh=hard --overwrite
application.argoproj.io/demo-app annotated
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$
```

7 Centralisation des logs (Loki/Grafana)

7.01 Objectifs :

Mettre en place une centralisation des logs Kubernetes avec :
Loki
Grafana
Promtail

7.02 Résultats :

L'intégration complète n'a pas été finalisée dans l'environnement local k3d/k3s en raison de problèmes de compatibilité et de configuration entre Loki et Grafana.
Ce qui a été testé :
déploiement Helm de Loki
datasource Grafana
collecte des logs pods
tests d'ingestion
Difficultés rencontrées
incompatibilités de versions
datasource non fonctionnelle
problèmes d'ingestion des logs
limitations environnement local k3d

7.1 Architecture Loki

Pods
↓
stdout/stderr
↓
Promtail
↓
Loki
↓

Grafana

7.2 Installation

helm install loki grafana/loki \
 --namespace logging \
 --create-namespace \
 --set deploymentMode=SingleBinary \
 --set singleBinary.replicas=1 \
 --set backend.replicas=0 \
 --set read.replicas=0 \
 --set write.replicas=0 \
 --set gateway.enabled=false \
 --set lokiCanary.enabled=false \
 --set test.enabled=false \
 --set loki.auth_enabled=false \
 --set loki.useTestSchema=true \
 --set minio.enabled=true \
 --set loki.storage.bucketNames.chunks=chunks \
 --set loki.storage.bucketNames.ruler=ruler \
 --set loki.storage.bucketNames.admin=admin

```
henri@Henri:~/labs/sre-kubernetes-lab$ helm install loki grafana/loki \
  --namespace logging \
  --create-namespace \
  --set deploymentMode=SingleBinary \
  --set singleBinary.replicas=1 \
  --set backend.replicas=0 \
  --set read.replicas=0 \
  --set write.replicas=0 \
  --set gateway.enabled=false \
  --set lokiCanary.enabled=false \
  --set test.enabled=false \
  --set loki.auth_enabled=false \
  --set loki.useTestSchema=true \
  --set minio.enabled=true \
  --set loki.storage.bucketNames.chunks=chunks \
  --set loki.storage.bucketNames.ruler=ruler \
  --set loki.storage.bucketNames.admin=admin
```

A l afin de l'installation , on vosi le message : qui donne : http://loki.logging.svc.cluster.local:3100/

```
curl -H "Content-Type: application/json" -XPOST -s "http://127.0.0.1:3100/loki/api/v1/push"  \
--data-raw "{\"streams\": [{\"stream\": {\"job\": \"test\"}, \"values\": [[\"$(date +%s)000000000\", \"fizzbuzz\"]]}]}"
```

Then verify that Loki did receive the data using the following command:

```
curl "http://127.0.0.1:3100/loki/api/v1/query_range" --data-urlencode 'query={job="test"}' | jq .data.result
```

```
***************************************************************************
Connecting Grafana to Loki
***************************************************************************

If Grafana operates within the cluster, you'll set up a new Loki datasource by utilizing the following URL:

http://loki.logging.svc.cluster.local:3100/
henri@Henri:~/labs/sre-kubernetes-lab$
```

7.5 Vérifications

kubectl get pods -n logging

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n logging
NAME                   READY   STATUS    RESTARTS   AGE
loki-0                 2/2     Running   0          8m9s
loki-chunks-cache-0    2/2     Running   0          8m9s
loki-minio-0           1/1     Running   0          8m9s
loki-results-cache-0   2/2     Running   0          8m9s
```

7.6 logging/promtail-values.yaml

helm install promtail grafana/promtail \
 --namespace logging \
 --set "config.clients[0].url=http://loki:3100/loki/api/v1/push"

```
henri@Henri:~/labs/sre-kubernetes-lab$ helm install promtail grafana/promtail \
  --namespace logging \
  --set "config.clients[0].url=http://loki:3100/loki/api/v1/push"
WARNING: This chart is deprecated
NAME: promtail
LAST DEPLOYED: Sat May  9 16:48:42 2026
NAMESPACE: logging
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
**********************************************************************
 Welcome to Grafana Promtail
 Chart version: 6.17.1
 Promtail version: 3.5.1
**********************************************************************

Verify the application is working by running these commands:
* kubectl --namespace logging port-forward daemonset/promtail 3101
* curl http://127.0.0.1:3101/metrics
henri@Henri:~/labs/sre-kubernetes-lab$
```

Vérifier promtail

kubectl --namespace logging port-forward daemonset/promtail 3101
puis :
curl http://127.0.0.1:3101/metrics

Vérifier : kubectl get pods -n logging

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n logging
NAME                              READY   STATUS    RESTARTS   AGE
loki-0                            2/2     Running   0          20m
loki-canary-9c475                 1/1     Running   0          20m
loki-canary-krk22                 1/1     Running   0          20m
loki-canary-zqq2f                 1/1     Running   0          20m
loki-gateway-7bdf4bbf46-fp698     1/1     Running   0          20m
loki-minio-0                      1/1     Running   0          20m
promtail-hmtz8                    1/1     Running   0          57s
promtail-nj674                    1/1     Running   0          57s
promtail-vpz69                    1/1     Running   0          57s
henri@Henri:~/labs/sre-kubernetes-lab$
```

Vérifier les logs promtail

7.8 Générer des logs

Créer le fichier : /logging/log-generator.yaml

```
apiVersion: apps/v1
kind: Deployment

metadata:
  name: log-generator
  namespace: demo-app

spec:
  replicas: 1

  selector:
    matchLabels:
      app: log-generator

  template:
    metadata:
      labels:
        app: log-generator

    spec:
      containers:
      - name: log-generator
        image: busybox

        command:
        - /bin/sh
        - -c
        - |
          while true; do
            echo "$(date) INFO Application running"
            sleep 2
          done
```

7.9 Déployer

kubectl apply -f logging/log-generator.yaml

7.10 Vérifier les log K8s

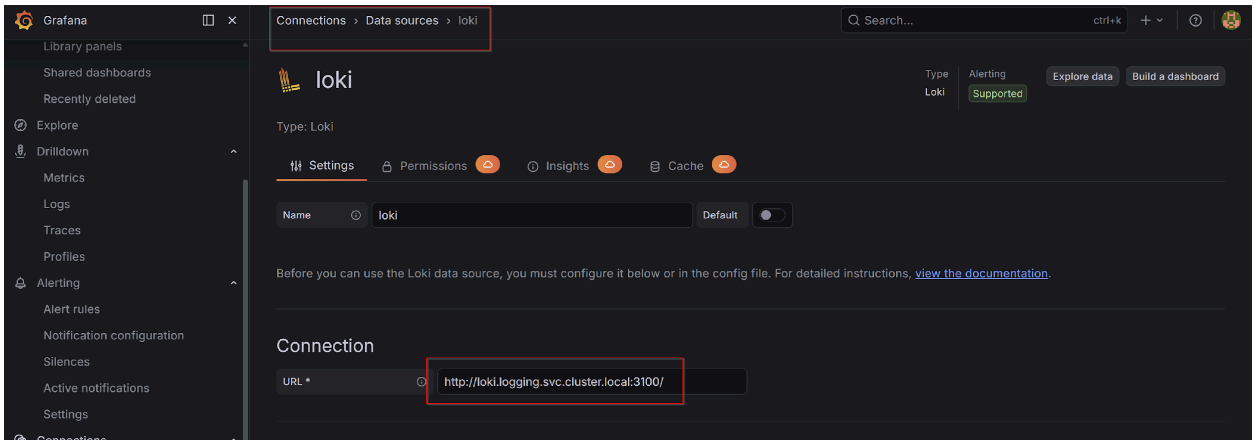
kubectl logs -n demo-app deployment/log-generator

On devrais voir : INFO Application running

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl logs -n demo-app deployment/log-generator
Sat May  9 01:04:53 UTC 2026 INFO Application running
Sat May  9 01:04:55 UTC 2026 INFO Application running
Sat May  9 01:04:57 UTC 2026 INFO Application running
Sat May  9 01:04:59 UTC 2026 INFO Application running
```

7.11 Vérifier que Loki reçoit les logs

Aller dans Grafana
Ajouter se ce n'est déjà fait :
Connections
→ Data Sources
→ Add Data Source
→ Loki
Saisir l'URL de Loki : http://loki.logging:3100 ou bien **http://loki-gateway.logging.svc.cluster.local**



Puis pour continuer les tests : :

kubectl port-forward svc/<service-name> -n logging 3100:3100
Dans notre lab service-name = loki
Donc :
kubectl port-forward svc/loki -n logging 3100:3100

puis :
curl http://localhost:3100/ready

si on voit eady . c'Est OK

7.12 Tester les logs dans Grafana

Dans : Explore
Requête : {job=~".+"}

requête
ou :
{app="log-generator"}
Ou :
Logs Traefik

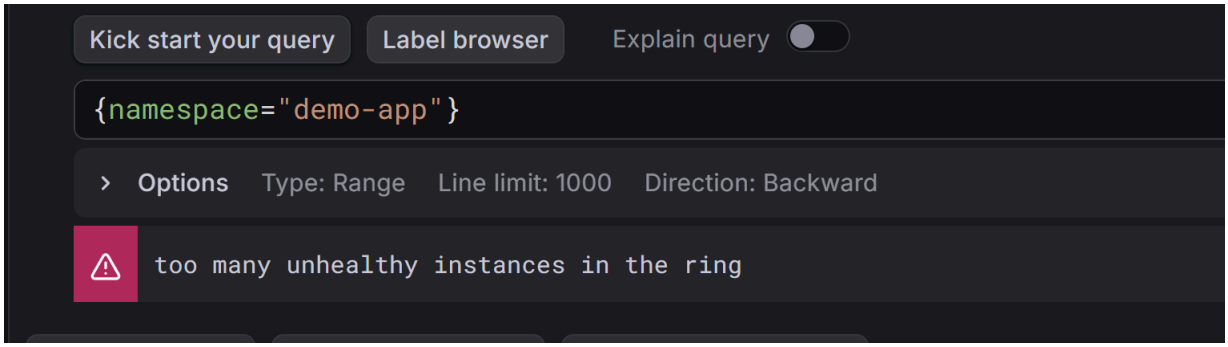
```
{namespace="kube-system"}
```

7.13 Erreur frequente avec Grafana/loki

J'Ai constaté qu'on a e message de waring : ''too many unhealthy instance in the ring, dans Grafana et qu'on utilise la requête : : {namespace="demo-app"}
Solution :
Supprimer le pod loki : kubectl delete pod loki-0 -n logging
Attendre, Le StatefulSet va le recréer automatiquement.
kubectl get pods -n logging -w

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n logging -w
NAME                           READY   STATUS    RESTARTS   AGE
loki-0                         2/2     Running   0          54s
```

Verifier que loki-0 2/2 running



---

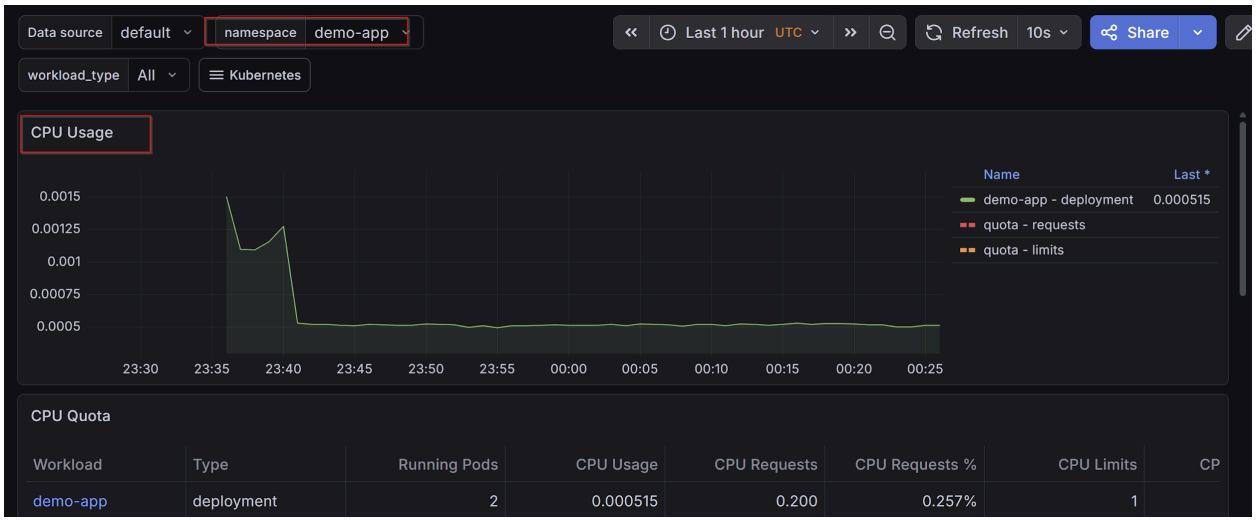
8 État du cluster au repos (avant charge)

8.1 État du cluster avant les tests de charge

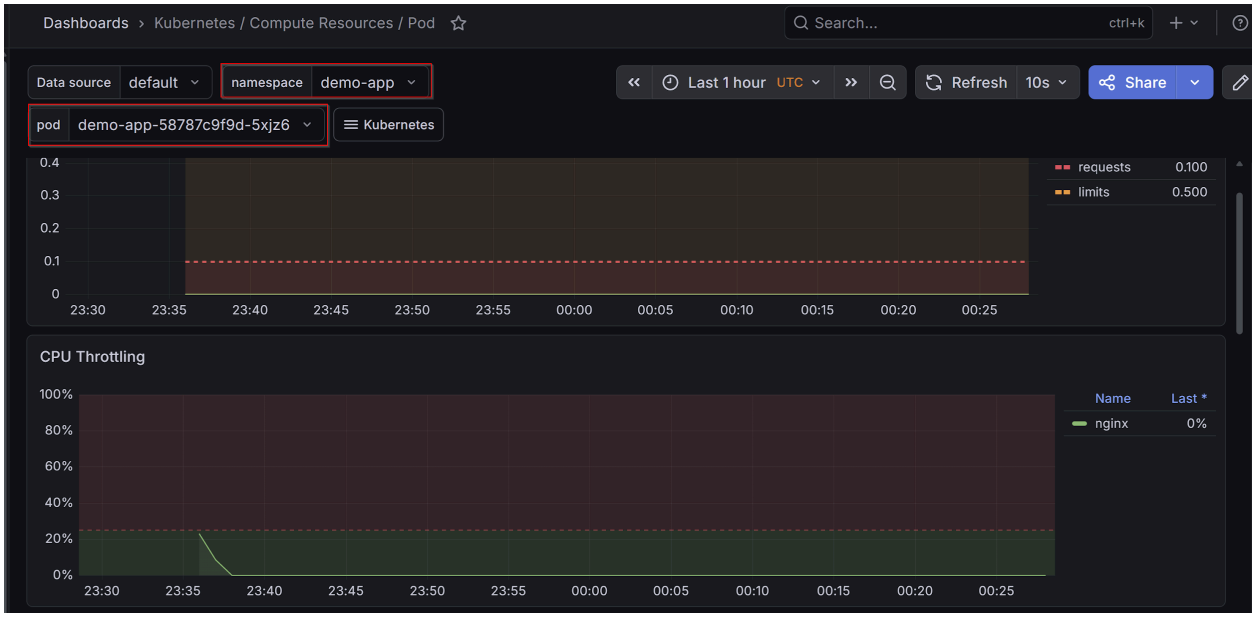
Il n'y a au départ que 2 pods dans le cluster

```
henri@Henri:~/labs/sre-kubernetes-lab$
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n demo-app
NAME                        READY   STATUS    RESTARTS   AGE
demo-app-58787c9f9d-5xjz6   1/1     Running   0          55m
demo-app-58787c9f9d-vtqgd   1/1     Running   0          55m
henri@Henri:~/labs/sre-kubernetes-lab$
```

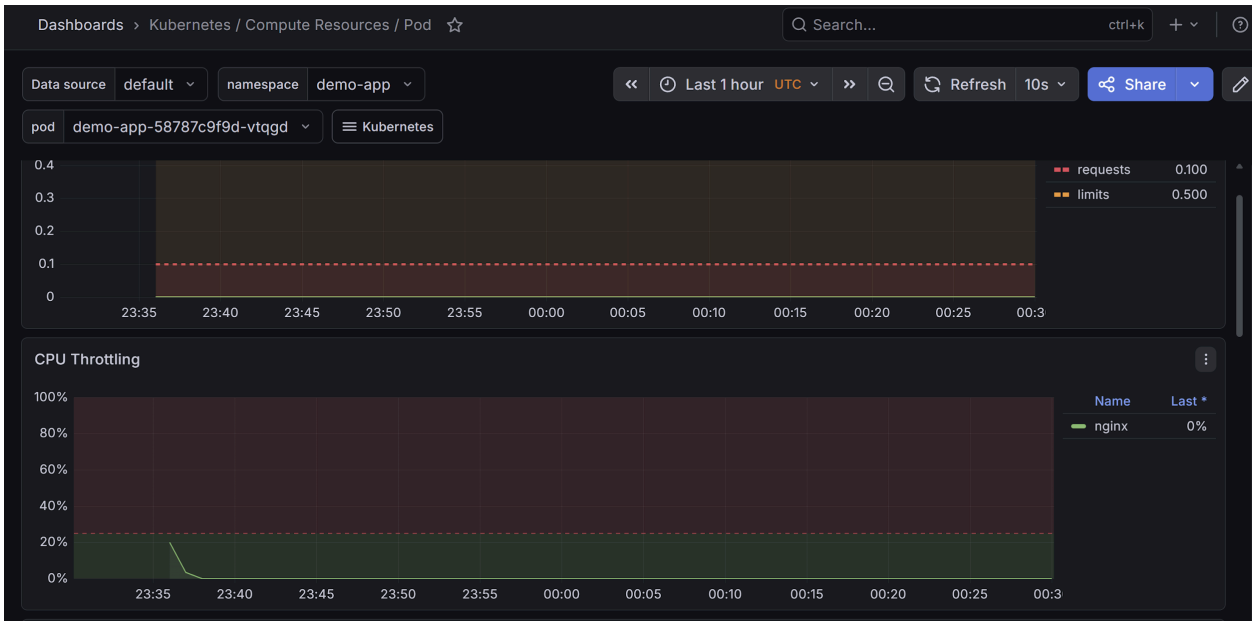
Charge CPU sur le cluster : demo-app :



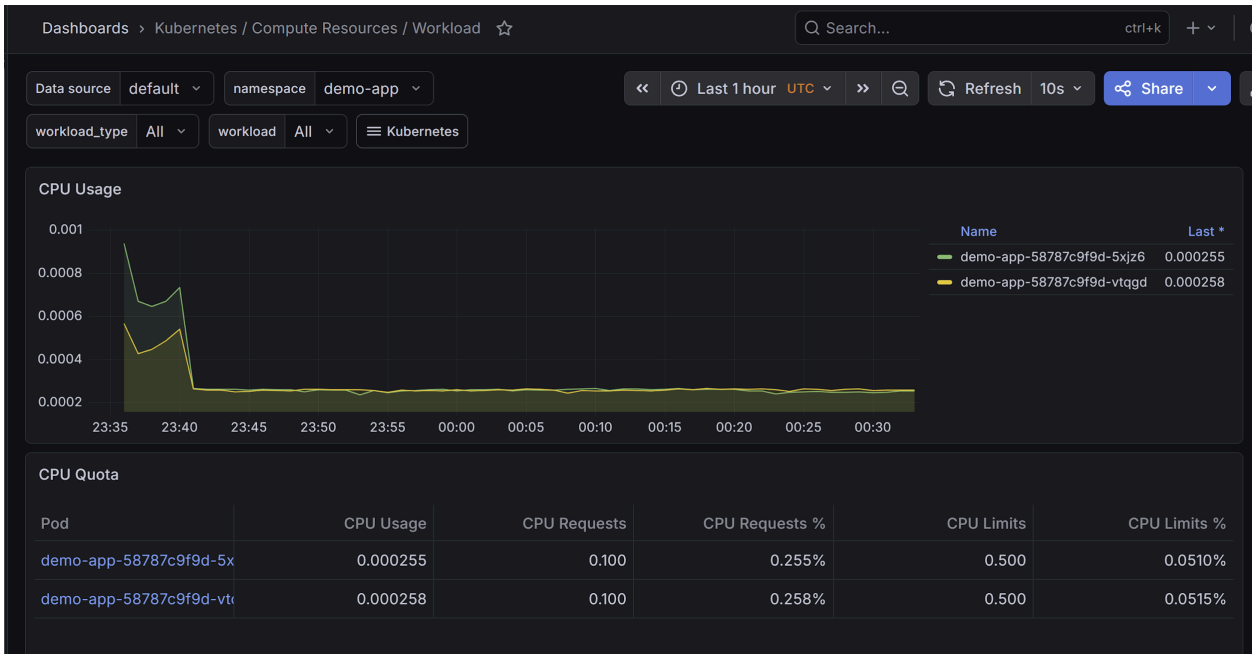
Charge CPU sur le pod 1 :



Charge CPU sur le pod 2 :

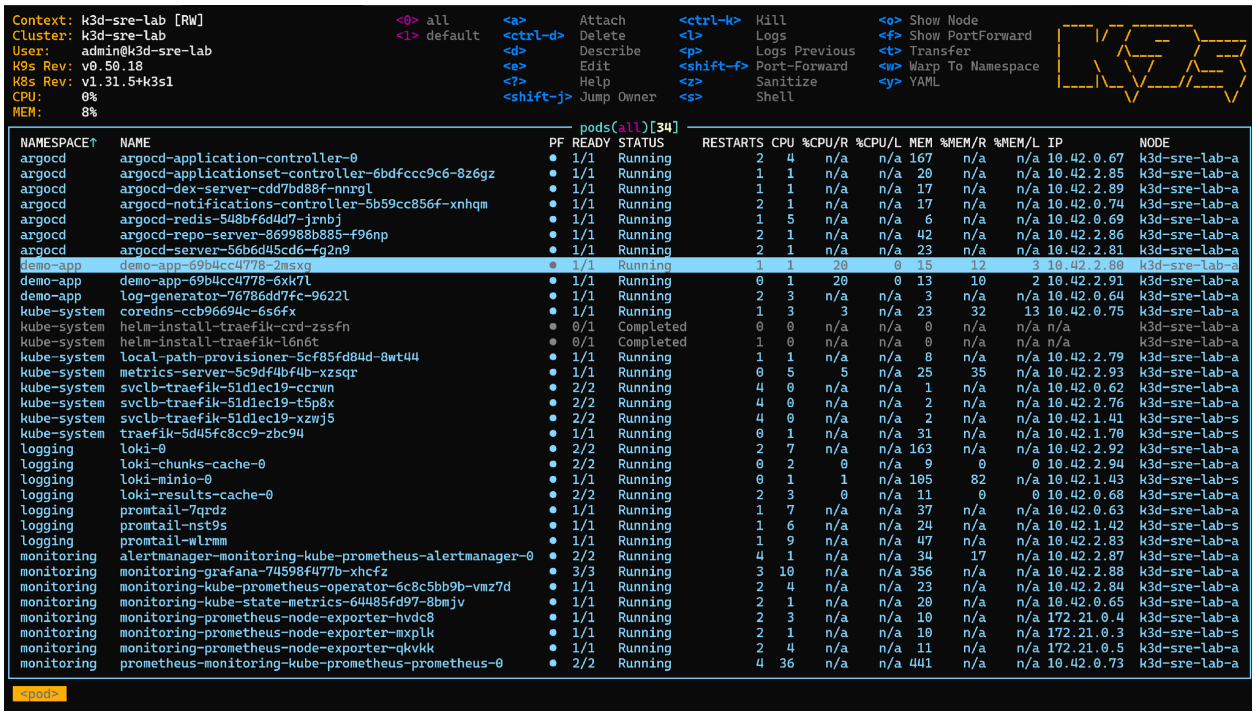


Pour l'ensemble des 2 pods :



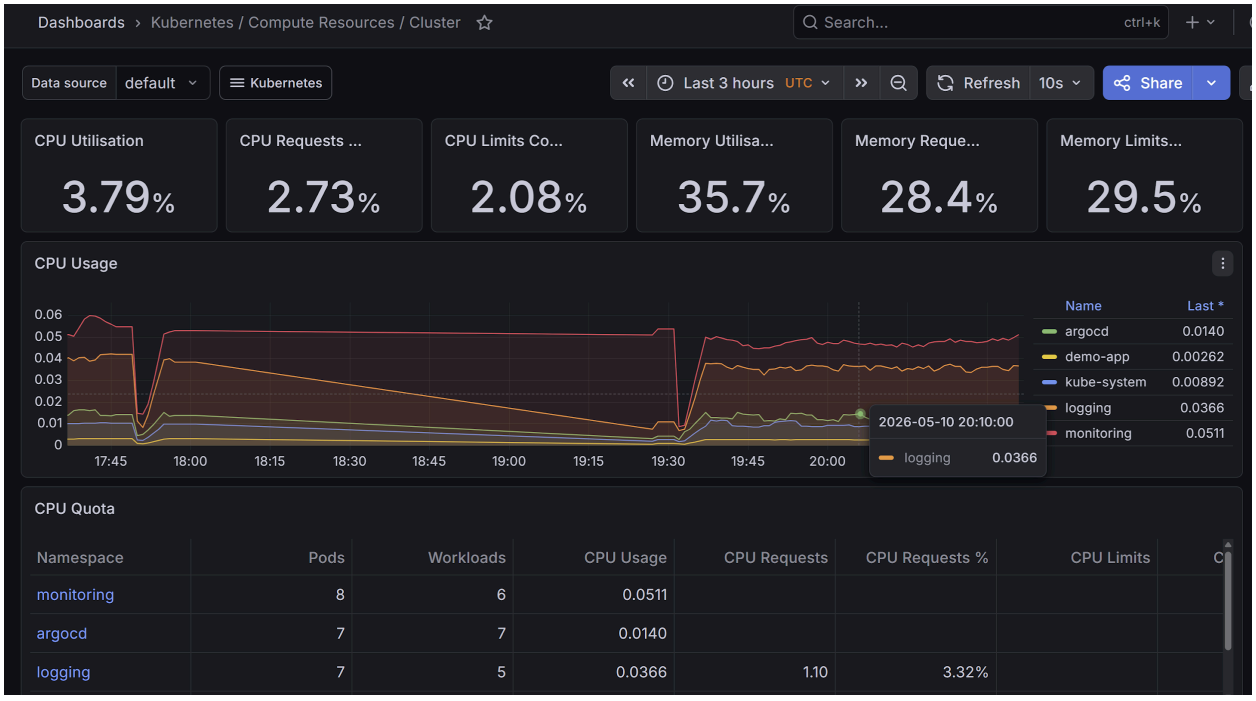
8.2 k9s – vue pods

Vue K9s
Vue de l'ensemble des pods



8.3 Grafana – Vue cluster

Vue k9s
Vue de l'ensemble des namespaces du cluster



9 Load Testing

9.1 Test 1 (K6)

9.1.1 Installer K6

```
henri@Henri:~/labs/sre-kubernetes-lab$ k6
Command 'k6' not found, but can be installed with:
sudo snap install k6
henri@Henri:~/labs/sre-kubernetes-lab$ sudo snap install k6
[sudo] password for henri:
2026-05-08T17:29:23-04:00 INFO Waiting for automatic snapd restart...
Automatically connect eligible plugs and slots of snap "snapd"
```

9.1.2 fichier de test

Load-testing/k6-loadtest.js

```
import http from 'k6/http';

export const options = {
  vus: 200,
  duration: '5m',
};

export default function () {
  http.get('http://localhost:8080', {
    headers: {
      Host: 'demo.local',
    },
  });
}
```

9.1.3 Run:

k6 run k6-loadtest.js

k6 run load-testing/k6-loadtest.js

```
henri@Henri:~/labs/sre-kubernetes-lab$ cd load-testing
henri@Henri:~/labs/sre-kubernetes-lab/load-testing$ k6 run k6-loadtest.js

         /\      Grafana   /‾‾/
    /\  /  \     |\  __   /  /
   /  \/    \    | |/ /  /   ‾‾\
  /          \   |   (  |  (‾)  |
 / __________ \  |_|\_\  \_____/


     execution: local
        script: k6-loadtest.js
        output: -

     scenarios: (100.00%) 1 scenario, 20 max VUs, 2m30s max duration (incl. graceful stop):
              * default: 20 looping VUs for 2m0s (gracefulStop: 30s)


running (0m09.0s), 20/20 VUs, 175 complete and 0 interrupted iterations
default   [=>----------------------------------------] 20 VUs  0m09.1s/2m0s
```

9.1.4 Observer scaling :

kubectl get hpa -w

au départ
kubectl get pods -n demo-app -w

```
henri@Henri:/mnt/c/Users/henri$ kubectl get pods -n demo-app -w
NAME                              READY   STATUS    RESTARTS       AGE
demo-app-69b4cc4778-2msxg         1/1     Running   1 (128m ago)   13h
demo-app-69b4cc4778-6xk7l         1/1     Running   0              12h
log-generator-76786dd7fc-9622l    1/1     Running   2 (128m ago)   39h
```

Puis en cours de charge K6 :

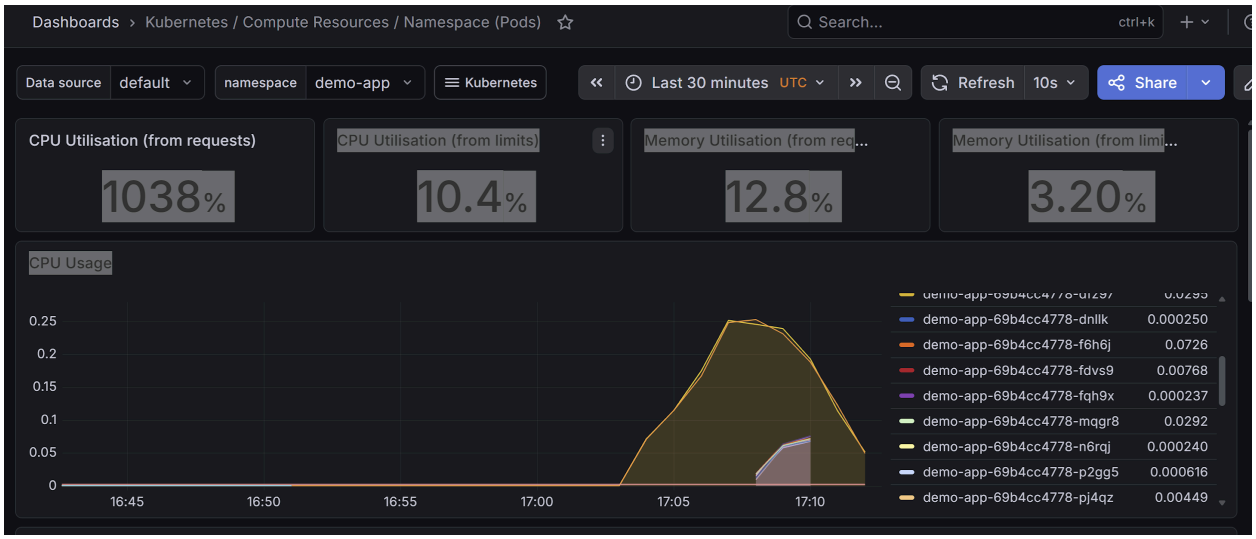
Et aussi :

```
henri@Henri:/mnt/c/Users/henri$ kubectl get hpa -w -n demo-app
NAME           REFERENCE               TARGETS         MINPODS   MAXPODS   REPLICAS   AGE
demo-app-hpa   Deployment/demo-app   cpu: 20%/50%    2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 4500%/50%   2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 10270%/50%   2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 8370%/50%    2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 8370%/50%    2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 20%/50%      2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 20%/50%      2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 1040%/50%    2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 5640%/50%    2         10        6          41h
demo-app-hpa   Deployment/demo-app   cpu: 9990%/50%    2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 3070%/50%    2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 20%/50%      2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 6820%/50%    2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 7320%/50%    2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 10760%/50%   2         10        2          41h
demo-app-hpa   Deployment/demo-app   cpu: 10910%/50%   2         10        6          41h
demo-app-hpa   Deployment/demo-app   cpu: 9970%/50%    2         10        10         41h
demo-app-hpa   Deployment/demo-app   cpu: 6846%/50%    2         10        10         41h
demo-app-hpa   Deployment/demo-app   cpu: 3440%/50%    2         10        2          41h
```

verifier dans Grafana :

<u>Grafana sur les dernières 30 minutes :</u>

Dasn Grafana lors du test k6 , on constate un pic a 0.25 de charge CPU. Soit 25% charge CPU



Analyse des résultats :

✅ forte montée CPU
✅ réaction du HPA
✅ scale up jusqu'à 10 pods
✅ scale down ensuite
✅ métriques Grafana cohérentes

Graphe CPU Usage de Gafana , Le pic : `0.25 en ordonnées du graphe` signifie :
$0.25 \times 100 = 25\%\ CPU\ re'el$ Donc :un quart de CPU complet utilisé , ce qui est cohérent avec un test k6 agressif
Pourquoi 1038% paraît énorme Parce que : requests: cpu: 25m, probablement. Donc :SI consommation réelle :250m alors : $\frac{250m}{25m} \times 100 = 1000\%$
exactement ce que Grafana affiche.

<u>Grafana sur la dernière heure :</u>

9.2 Test 2 (hey)

kubectl port-forward svc/demo-app 8888:80 -n demo-app

hey -z 5m -c 500 http://localhost:8888

-z 5m : durée du test = 5 minutes
-c 500 : nombre de comnectins simultanées

On observe alors que HPA fonctionne , le cluster scale de 2 à 10 replicas. On passe de 2 à 10 pods
Cf figure ci-dessous.

```
henri@Henri:/mnt/c/Users/henri$ kubectl get pods -n demo-app -w
NAME                              READY   STATUS    RESTARTS      AGE
demo-app-69b4cc4778-548wv         1/1     Running   0             2m7s
demo-app-69b4cc4778-6mfpj         1/1     Running   0             114s
demo-app-69b4cc4778-94twr         1/1     Running   0             114s
demo-app-69b4cc4778-gmvtd         1/1     Running   0             2m7s
demo-app-69b4cc4778-hf74z         1/1     Running   0             99s
demo-app-69b4cc4778-k6ql2         1/1     Running   0             114s
demo-app-69b4cc4778-lvvgb         1/1     Running   0             114s
demo-app-69b4cc4778-q9sbd         1/1     Running   0             10m
demo-app-69b4cc4778-qdxkq         1/1     Running   0             99s
demo-app-69b4cc4778-vswwg         1/1     Running   0             7m4s
log-generator-76786dd7fc-9622l    1/1     Running   1 (11h ago)   26h
```

Resultats du test hey

```
henri@Henri:/mnt/c/Users/henri$ hey -z 5m -c 500 http://localhost:8888

Summary:
  Total:        300.0701 secs
  Slowest:      0.5667 secs
  Fastest:      0.0014 secs
  Average:      0.1500 secs
  Requests/sec: 7788.6412

  Total data:   2094075648 bytes
  Size/request: 2094 bytes

Response time histogram:
  0.001 [1]     |
  0.058 [409240]        |■■■■■■■■■■■■■■■■■■■■■■■■■■■■
  0.114 [571740]        |■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
  0.171 [18113] |■
  0.228 [548]   |
  0.284 [16]    |
  0.341 [67]    |
  0.397 [75]    |
  0.454 [62]    |
  0.510 [130]   |
  0.567 [8]     |
```

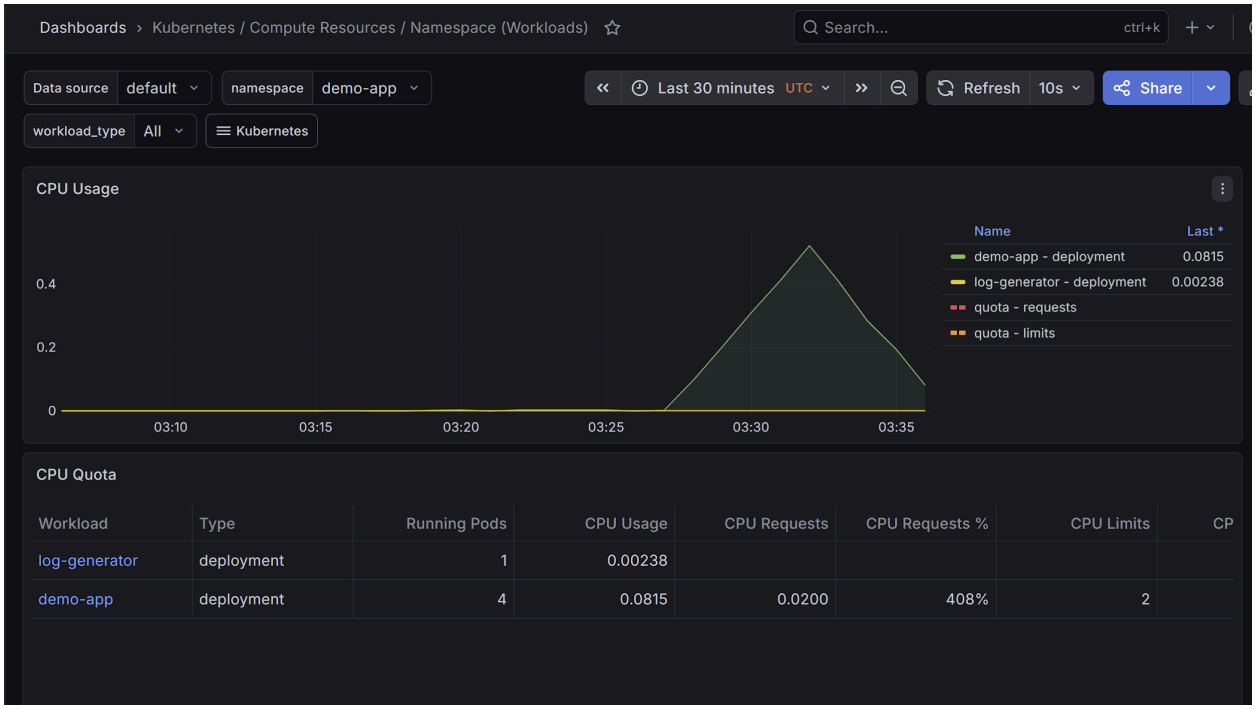
```
Latency distribution:
  10% in 0.0306 secs
  25% in 0.0375 secs
  50% in 0.0679 secs
  75% in 0.0852 secs
  90% in 0.0982 secs
  95% in 0.1054 secs
  99% in 0.1206 secs

Details (average, fastest, slowest):
  DNS+dialup:   0.0000 secs, 0.0014 secs, 0.5667 secs
  DNS-lookup:   0.0000 secs, 0.0000 secs, 0.0461 secs
  req write:    0.0000 secs, 0.0000 secs, 0.0101 secs
  resp wait:    0.1463 secs, 0.0013 secs, 0.5163 secs
  resp read:    0.0036 secs, 0.0000 secs, 0.1415 secs

Status code distribution:
  [200] 1000000 responses
```

Résutats dans Grafana :

Le test de charge avec hey a durée 5 minutes et pendant ce laps de temps la charge CPU pour le cluster (les 2 pods qui tournent dans demo-app)) est monté à 0.4, puis elle est retombée à 0.1 a la fin du test



Le nombre de pods a scalé, il est passé de 2 à 10 .

L'utilisation CPU pour chaque pod du namespace est représenté dans la figure ci-dessous.

On constate dans le graphe Grafana, un pic lors de la durée du test, pour lequel on voit que le CPU a grimpé jusqu'à 0.4

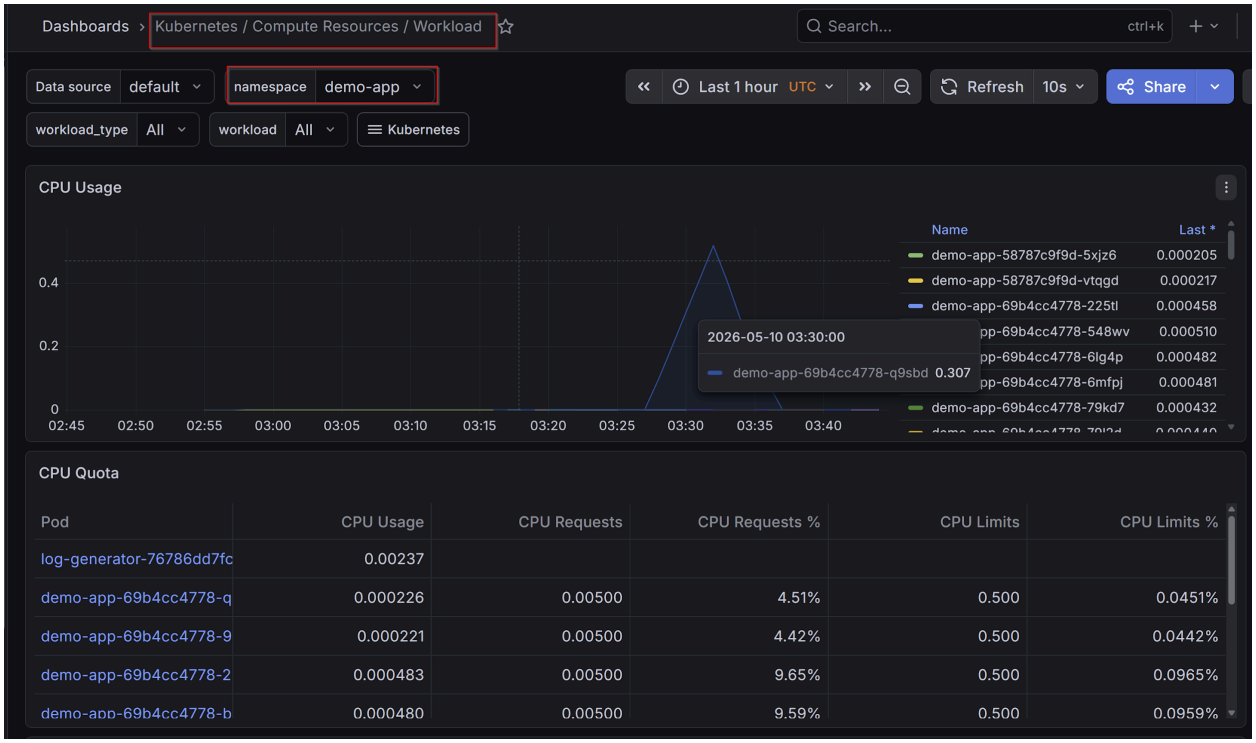
Rappel : 0.4 CPU cores=400m (400 miliCPU) c'est-à-dire : 40% d'un cœur CPU ou 400 milliCPU.

Donc la charge est montée jusqu'à 40 % , puis est retombée à 1m (soit 0.1%)

Ce qu'on constate également avec la commande kubectl top nodes, dans la figure ci-dessous , dans laquelle on voit les 10 pods qui tournent chacun avec seulement 1m (0.1%)

kubectl top pods -n demo-app

```
henri@Henri:/mnt/c/Users/henri$ kubectl top pods -n demo-app
NAME                                CPU(cores)   MEMORY(bytes)
demo-app-69b4cc4778-2msxg           1m           13Mi
demo-app-69b4cc4778-4twh5           1m           13Mi
demo-app-69b4cc4778-94twr           1m           13Mi
demo-app-69b4cc4778-dnllk           1m           13Mi
demo-app-69b4cc4778-fqh9x           1m           13Mi
demo-app-69b4cc4778-n6rqj           1m           13Mi
demo-app-69b4cc4778-q9sbd           1m           13Mi
demo-app-69b4cc4778-shvvc           1m           13Mi
demo-app-69b4cc4778-sjm2t           1m           13Mi
demo-app-69b4cc4778-xzl57           1m           13Mi
log-generator-76786dd7fc-9622l      3m           3Mi
henri@Henri:/mnt/c/Users/henri$
```



9.3 Test K6 plus agressif

```
import http from 'k6/http';

export const options = {
  stages: [
    { duration: '1m', target: 100 },
    { duration: '2m', target: 500 },
    { duration: '2m', target: 1000 },
    { duration: '1m', target: 0 },
```

```
  ],
};

export default function () {
  http.get('http://localhost:8080', {
    headers: {
      Host: 'demo.local',
    },
  });
}
```

9.4 Tests Logs Lki

Observer les logs dans Grafana suite aux tests 1 & 2 précédents
Si on arrive a faire fonctionner Loki …!!!!
Ce qui n'a pas été le cas dans ce lab.

---

10 Incident Simulation

10.1 Ingress Failure

On teste la résilience du système. La destruction d'un pod , entraine sa recréation immédiate
On veut tester la résilience d'un pod traefik.

kubectl get pod -n kube-system.

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n kube-system -w
NAME                                      READY   STATUS      RESTARTS        AGE
coredns-ccb96694c-6s6fx                   1/1     Running     1 (4h49m ago)   41h
helm-install-traefik-crd-zssfn            0/1     Completed   0               2d3h
helm-install-traefik-l6n6t                0/1     Completed   1               2d3h
local-path-provisioner-5cf85fd84d-8wt44   1/1     Running     1 (4h49m ago)   41h
metrics-server-5c9df4bf4b-xzsqr           1/1     Running     0               15h
svclb-traefik-51d1ec19-ccrwn              2/2     Running     4 (4h49m ago)   2d3h
svclb-traefik-51d1ec19-t5p8x              2/2     Running     4 (4h49m ago)   2d3h
svclb-traefik-51d1ec19-xzwj5              2/2     Running     4 (4h49m ago)   2d3h
traefik-5d45fc8cc9-j78bk                  1/1     Running     0               12s
```

Puis :

kubectl delete pod traefik-5d45fc8cc9-j78bk -n kube-system

on constate : que le pod traefik a été recrée par le système.

Le pod traefik-5d45fc8cc9-zbc94 a été recrée. Pod identique , mis à part le nom

```
henri@Henri:~/labs/sre-kubernetes-lab$ kubectl get pods -n kube-system -w
NAME                                      READY   STATUS      RESTARTS       AGE
coredns-ccb96694c-6s6fx                   1/1     Running     1 (5h1m ago)   41h
helm-install-traefik-crd-zssfn            0/1     Completed   0              2d4h
helm-install-traefik-l6n6t                0/1     Completed   1              2d4h
local-path-provisioner-5cf85fd84d-8wt44   1/1     Running     1 (5h1m ago)   41h
metrics-server-5c9df4bf4b-xzsqr           1/1     Running     0              15h
svclb-traefik-51d1ec19-ccrwn              2/2     Running     4 (5h1m ago)   2d4h
svclb-traefik-51d1ec19-t5p8x              2/2     Running     4 (5h1m ago)   2d4h
svclb-traefik-51d1ec19-xzwj5              2/2     Running     4 (5h2m ago)   2d4h
traefik-5d45fc8cc9-zbc94                  1/1     Running     0              6s
```